z-score, p-value, one-tailed & two-tailed test, correlation, z-test, t-test,chi-square test,f-distribution

Unit-4

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Z-Scores and P-Values

Situation: Heights in a Room

- Imagine you're in a room with 100 people, and you know their average height is **5 feet 7 inches** with a standard deviation of **3 inches**. Now, let's say you're **6 feet tall** and you want to know how unusual your height is compared to the other people in the room.

Step 1: Z-Score — How Far You Are from Average

The **z-score** tells you how far you are from the **average** (mean) height in terms of **standard deviations**.

- A **z-score** of **0** means you are exactly average.
- A positive **z-score** means you're taller than average; a negative **z-score** means you're shorter.
- **Example**: If you are 6 feet tall (72 inches), and the average is 5 feet 7 inches (67 inches), your **z-score** is:

$$Z = \frac{\text{Your height} - \text{Average height}}{\text{Standard deviation}} = \frac{72 - 67}{3} = 1.67$$

1.67 standard deviations taller than the average person in the room.

P-value

Step 2: P-Value — How Rare Is Your Height?

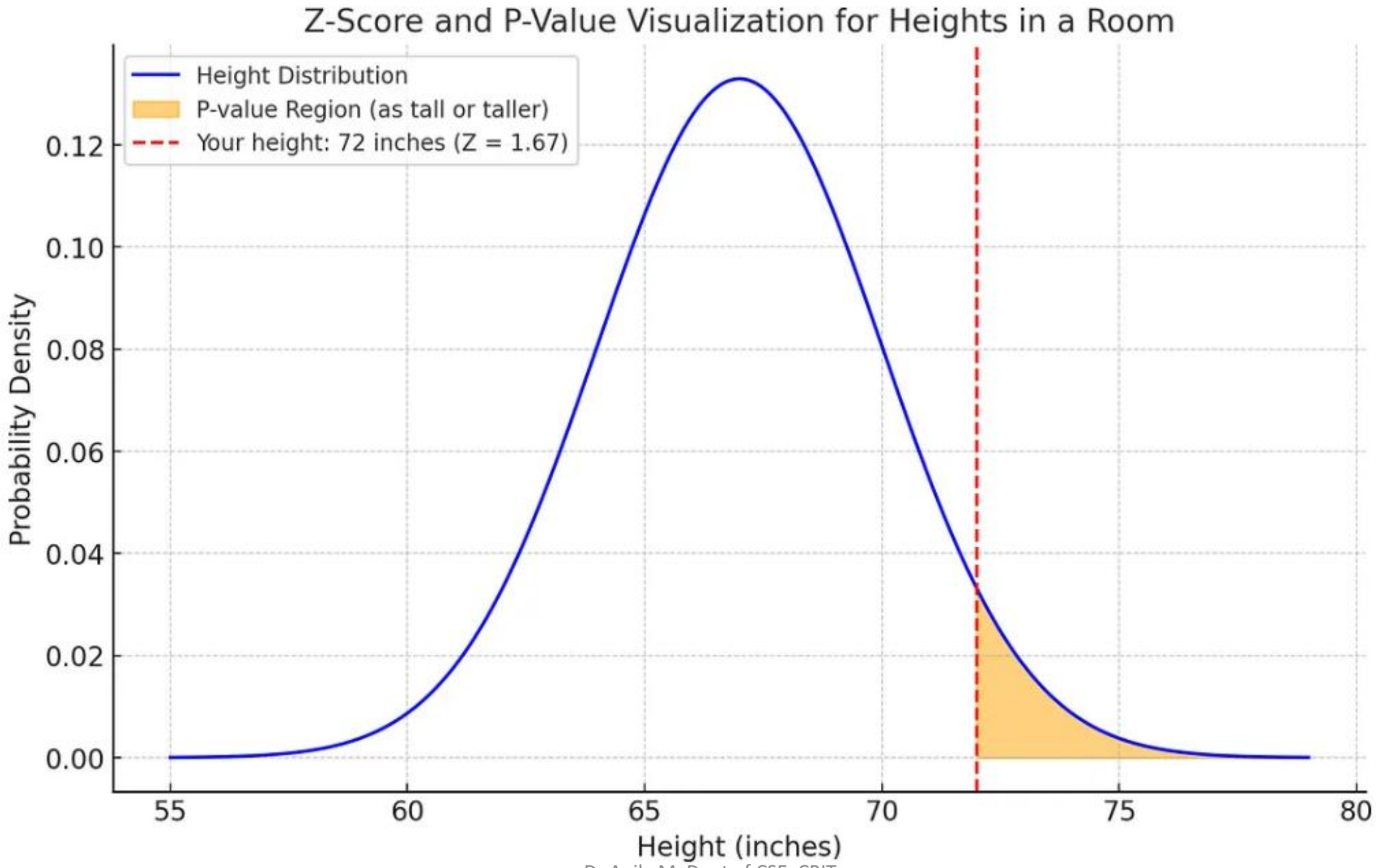
- The **p-value** tells you **how rare** or **common** it is to be as tall as you or taller. It's the probability of finding someone as tall (or taller) than you in this room.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817

P-Value= 1- 0.9525= 0.0475

•The p-value for a **z-score of 1.67** is around **0.0475** (or 4.75%).

•This means that **only 4.75%** of people in the room are as tall or taller than you.


Z-Score and P-Value Visualization for Heights in a Room
Height Distribution
P-value Region (as tall or taller)
Your height: 72 inches (Z = 1.67)
Probability Density
0.12
0.10
0.08
0.06
0.04
0.02
0.00
55
60
65
70
75
80
Height (inches)

Interrelationship Between Z-Score and P-Value

- The **z-score** measures how far you are from the average (how extreme your height is).
- The **p-value** measures **how likely** it is to find someone with a height as extreme as yours.
- As your **z-score** increases, meaning you get further from the average, the **p-value** gets smaller, meaning fewer people are as extreme as you.

Summary:

- **Z-score** is like saying, "I'm 1.67 standard deviations taller than the average."
- **P-value** is like saying, "Only 4.75% of the people in the room are as tall or taller than me."

Example: z-score, p-value calculation

- *A gardener believes that a new type of fertilizer will increase the average height of his tomato plants. Historically, his tomato plants have grown to an average height of 60 cm. He uses the new fertilizer on a sample of 25 plants and finds that their average height is 63 cm with a sample standard deviation of 5 cm.*

Z-score Calculation

Here:

- $\bar{x}$ = sample mean height = 63 cm
- μ = historical population mean height (our null hypothesis) = 60 cm
- s = sample standard deviation = 5 cm (since the population standard deviation is unknown, and our sample size is relatively small, we'd ideally use a t-test. However, for simplicity in this z-score example, we'll approximate using the sample standard deviation)
- n = sample size = 25

Calculating the z-score:

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{63 - 60}{\frac{5}{\sqrt{25}}} = \frac{3}{\frac{5}{5}} = \frac{3}{1} = 3$$

z-score of 3 indicates that the average height of the fertilized plants in our sample is 3 standard deviations higher than the historical average.

P-value Calculation

Our null hypothesis (H0) is that the new fertilizer has no effect on the average height (μ=60 cm). Our alternative hypothesis (Ha) is that the new fertilizer *increases* the average height (μ>60 cm) – this is a right-tailed test because the gardener is specifically interested in an increase in height.

To find the p-value for a right-tailed test with a z-score of 3, we need to find the area under the standard normal distribution curve to the right of z=3.

- Looking at a standard normal distribution table or using a calculator, the area to the left of z=3 is approximately 0.9987. Therefore, the area to the right of z=3 is:

$$p - value = P(Z > 3) = 1 - P(Z \leq 3) \approx 1 - 0.9987 = 0.0013$$

there is strong statistical evidence to suggest that the new fertilizer does indeed increase the average height of the tomato plants. The very low p-value indicates that observing a sample mean height of 63 cm (or even higher) would be very unlikely if the fertilizer had no effect on growth.

Application

- **z-scores and p-values are fundamental tools in statistical inference. They help us quantify the unusualness of our observations and make informed decisions about hypotheses based on data.**
- The z-score provides a standardized measure of how far our data is from the expected value, while the p-value gives us the probability of seeing such extreme data if our initial assumption (null hypothesis) were true

z-score (Standard Deviations)	p-value (Probability)	Confidence level
< -1.65 or > +1.65	< 0.10	90%
< -1.96 or > +1.96	< 0.05	95%
< -2.58 or > +2.58	< 0.01	99%

Real-time applications

Scientific Research: In almost every scientific discipline (biology, psychology, medicine, physics, etc.), researchers use z-scores (when population standard deviation is known or for large samples) and p-values to determine if their findings are statistically significant.

For example:

- Testing if a new drug is more effective than another.
- Comparing the average test scores of two different teaching methods.
- Investigating if there's a correlation between two variables.

Identifying Outliers: A z-score tells you how many standard deviations a particular data point is away from the mean.

- Data points with very high or very low z-scores (e.g., beyond ±2 or ±3) are often considered outliers, which can be important to identify for data cleaning or anomaly detection.

Market Research: Analyzing survey data to see if there are significant differences in opinions or preferences between different demographic groups

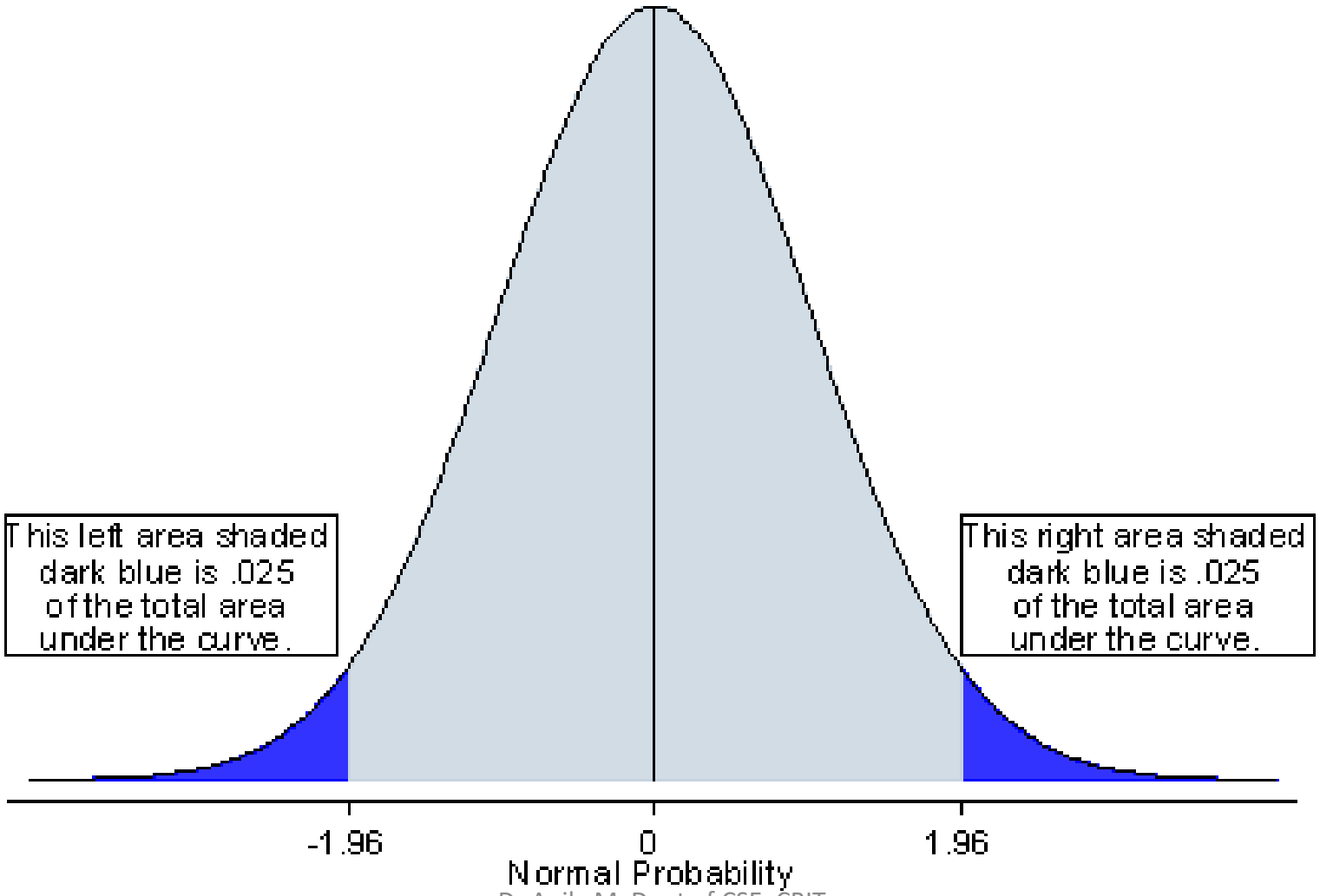
Scenario

- When you conduct a test of statistical significance, whether it is from a correlation, an ANOVA, a regression or some other kind of test, you are given a p-value somewhere in the output
- If your test statistic is symmetrically distributed, you can select one of three alternative hypotheses. Two of these correspond to one-tailed tests and one corresponds to a two-tailed test. However, the p-value presented is (almost always) for a two-tailed test.
- But how do you choose which test? Is the p-value appropriate for your test? And, if it is not, how can you calculate the correct p-value for your test given the p-value in your output?

two-tailed test

- If you are using a significance level of 0.05, a two-tailed test allots half of your alpha to test the statistical significance in one direction and half of your alpha to testing statistical significance in the other direction.
- This means that .025 is in each tail of the distribution of your test statistic. When using a two-tailed test, *regardless of the direction of the relationship you hypothesize*, you are testing for the possibility of the relationship in both directions.
- A two-tailed test will test both if the mean is significantly greater than *x* and less than *x*.
- The mean is considered significantly different from *x* if the test statistic is in the top 2.5% or bottom 2.5% of its probability distribution, resulting in a p-value less than 0.05.

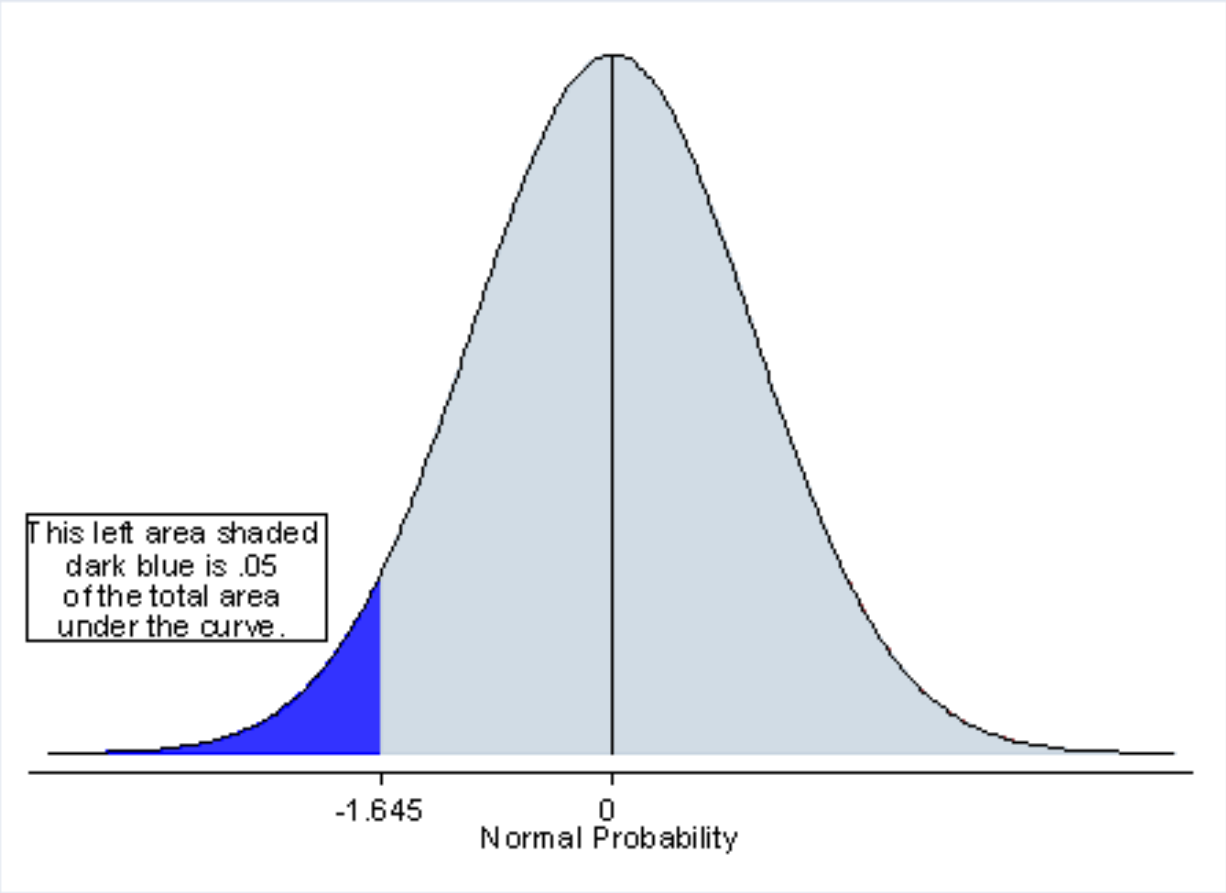
two-tailed test

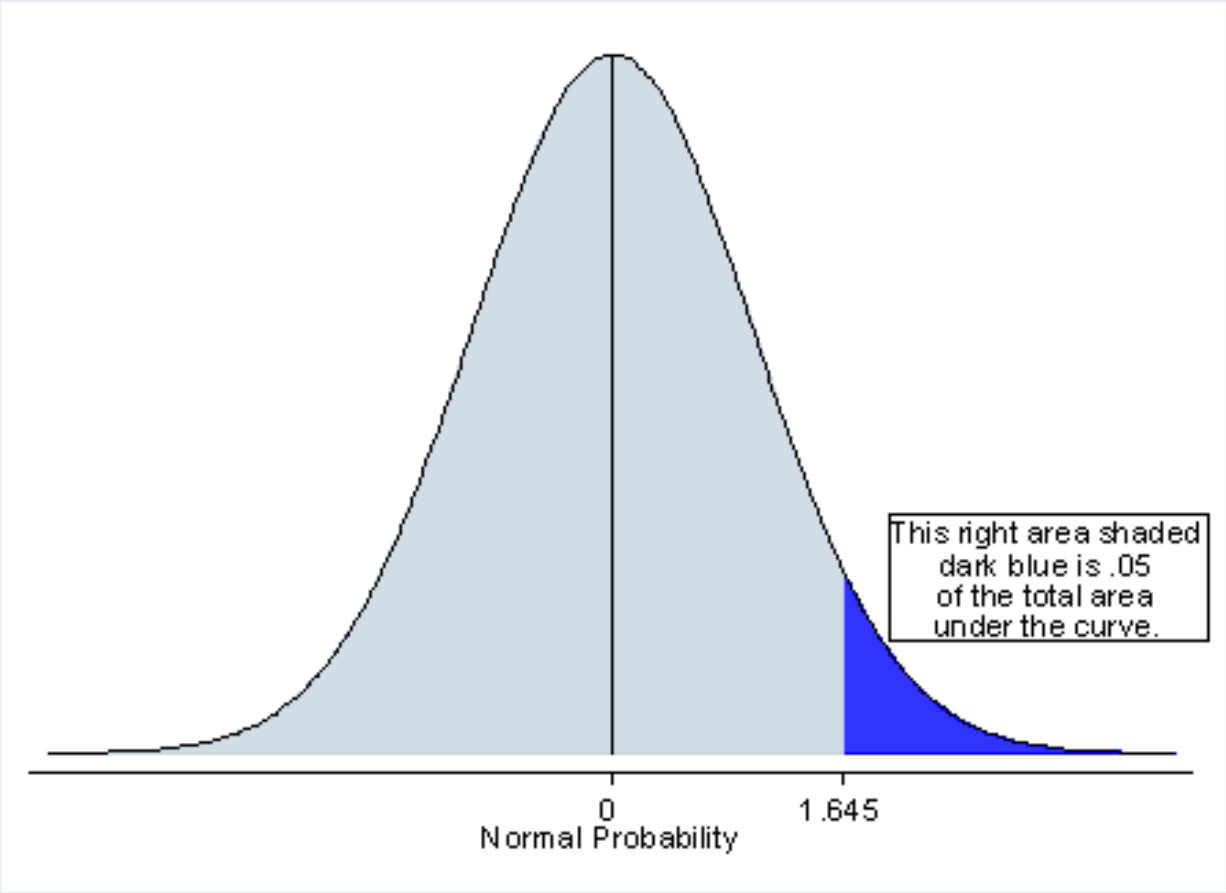


one-tailed test

- If you are using a significance level of .05, a one-tailed test allots all of your alpha to testing the statistical significance in the one direction of interest. This means that .05 is in one tail of the distribution of your test statistic. When using a one-tailed test, you are testing for the possibility of the relationship in one direction and completely disregarding the possibility of a relationship in the other direction.
- A one-tailed test will test either if the mean is significantly greater than *x* or if the mean is significantly less than *x*, but not both. Then, depending on the chosen tail, the mean is significantly greater than or less than *x* if the test statistic is in the top 5% of its probability distribution or bottom 5% of its probability distribution, resulting in a p-value less than 0.05

one-tailed test





Type 1 and Type-2 errors, Confidence interval

Statistical hypothesis testing implies that no test is ever 100% certain: that's because we rely on probabilities to experiment.

When online marketers and scientists run hypothesis tests, they're both looking for **statistically relevant results**. This means that the results of their tests have to be true within a range of probabilities (typically 95%).

Even though hypothesis tests are meant to be reliable, there are two types of errors that can still occur.

- These errors are known as **type 1 and type 2 errors** (or type i and type ii errors).

		Reality	
		True	False
Measured or Perceived	True	Correct ☺	**Type 1 error** False Positive
	False	**Type 2 error** False Negative	Correct ☺

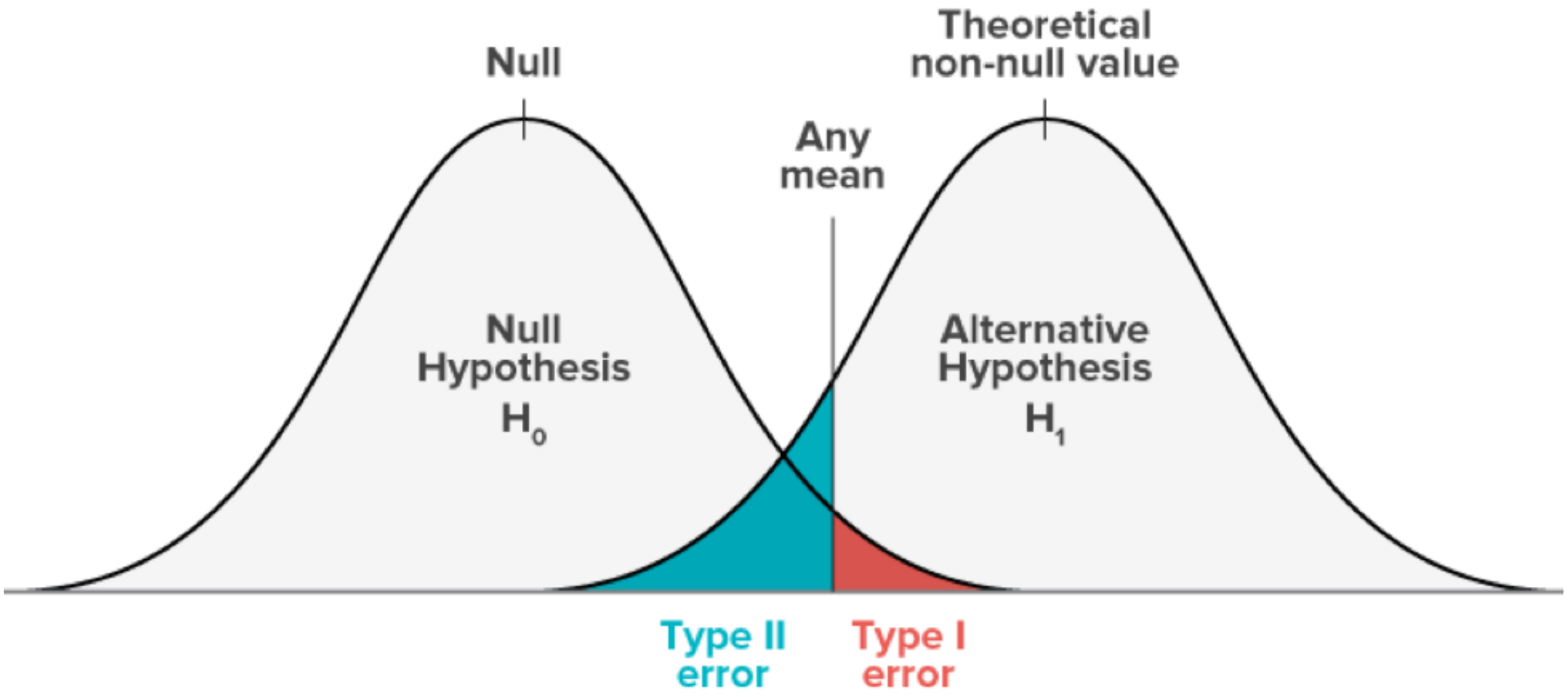
Type 1 errors

Type 1 errors – often assimilated with false positives – happen in hypothesis testing when the null hypothesis is true but rejected.

The null hypothesis is a general statement or default position that there is no relationship between two measured phenomena.

- Simply put, type 1 errors are “false positives” – they happen when the tester validates a statistically significant difference even though there isn’t one.

Type-I error



Type 1 errors have a probability of "α" correlated to the level of confidence that you set.
A test with a 95% confidence level means that there is a 5% chance of getting a type 1 error.

Why do type 1 errors occur?

Type 1 errors can happen due to bad luck (the 5% chance has played against you) or because you didn’t respect the test duration and sample size initially set for your experiment.

Consequently, a type 1 error will bring in a false positive. This means that you will wrongfully assume that your hypothesis testing has worked even though it hasn’t.

- In real-life situations, this could potentially mean losing possible sales due to a faulty assumption caused by the test.

Type II Errors

In the same way that type 1 errors are commonly referred to as “false positives”, type 2 errors are referred to as “false negatives”.

Type 2 errors happen when you inaccurately assume that no winner has been declared between a control version and a variation although there actually is a winner.

In more statistically accurate terms, type 2 errors happen when the null hypothesis is false and you subsequently fail to reject it.

- If the probability of making a type 1 error is determined by “α”, the probability of a type 2 error is “β”. Beta depends on the power of the test (i.e the probability of not committing a type 2 error, which is equal to 1-β).

Type II Errors

There are 3 parameters that can affect the power of a test:

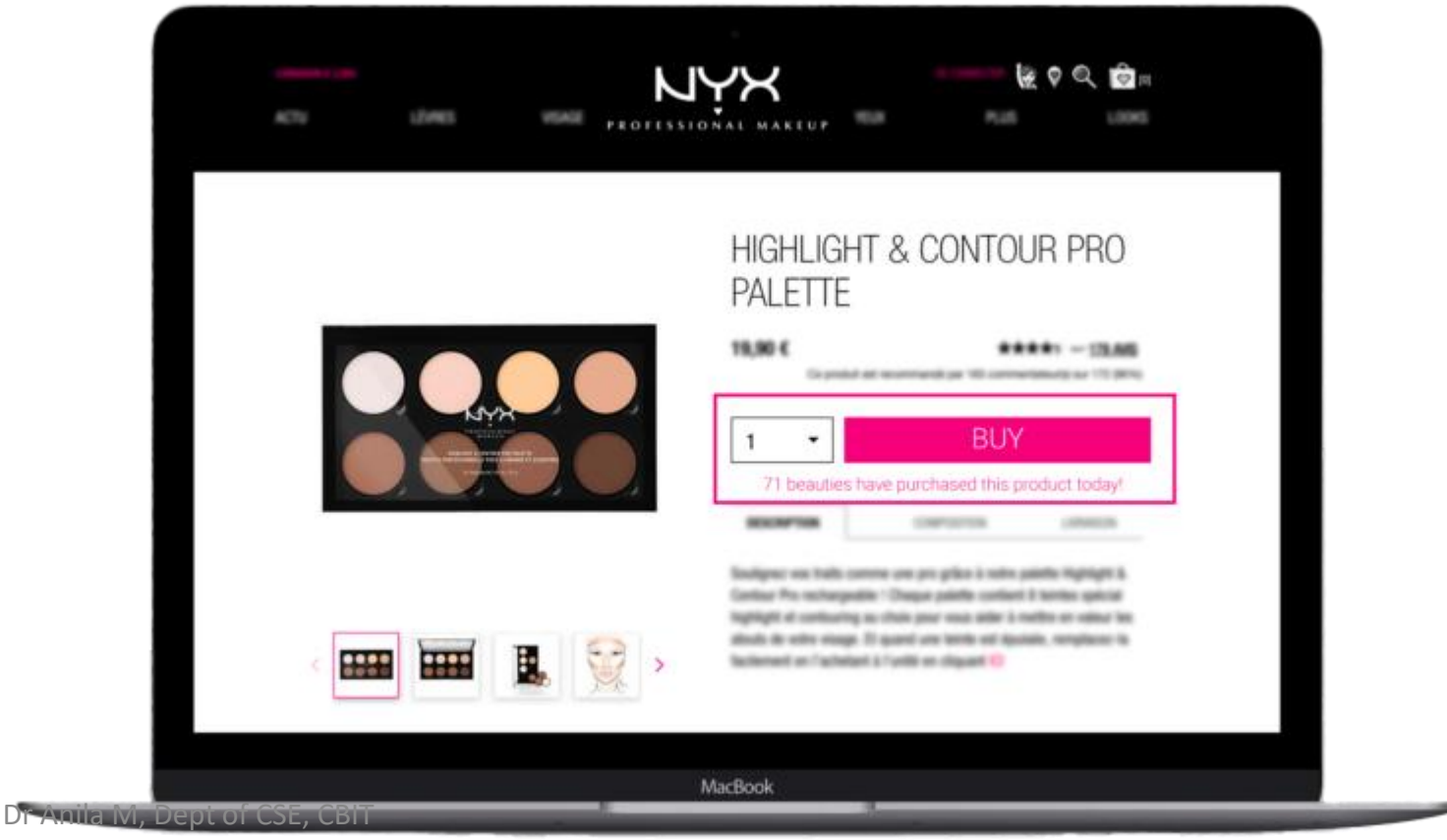
- Your sample size (n)
- The significance level of your test (α)
- The “true” value of your tested parameter

Similar to type 1 errors, type 2 errors can lead to false assumptions and poor decision-making that can result in lost sales or decreased profits.

- Moreover, getting a false negative (without realizing it) can discredit your conversion optimization efforts even though you could have proven your hypothesis.

Real-Life Example of a Type 2 Error

Let's say that you run an e-commerce store that sells cosmetic products for consumers. In an attempt to increase conversions, you have the idea to implement social proof messaging on your product pages, like **NYX Professional Makeup**.



Type-II error

You launch an A/B test to see if the variation (B) could outperform your control version (A).

After a week, you do not notice any difference in conversions: both versions seem to convert at the same rate and you start questioning your assumption. Three days later, you stop the test and keep your product page as it is.

At this point, you assume that adding social proof messaging to your store didn't have any effect on conversions.

Two weeks later, you hear that a competitor had added social proof messages at the same time and observed tangible gains in conversions. You decide to re-run the test for a month in order to get more statistically relevant results based on an increased level of confidence (say 95%).

After a month – surprise – you discover positive gains in conversions for the variation (B). Adding social proof messages under the purchase buttons on your product pages has indeed brought your company more sales than the control version.

- That's right – your first test encountered a **type 2 error!**

NOTE

Type one and type two errors are errors that we may encounter on a daily basis. It’s important to understand these errors and the impact that they can have on your daily life.

With type 1 errors you are making an incorrect assumption and can lose time and resources. Type 2 errors can result in a missed opportunity to change, enhance, and innovate a project.

- To avoid these errors, it’s important to pay close attention to the sample size and the significance level in each experiment.

T-test vs. Z-test

- Use t-tests when dealing with small samples or unknown variance, and Z-tests when samples are large and variance is known.
- A concept called **hypothesis testing**, along with several tests, including t-tests and z-tests, are some of the commonly used tools in analytics to establish relationships between data points.
- **Use a t-test:** When the sample size is small ($n < 30$) and/or the population variance is unknown.
- **Use a Z-test:** When the sample size is large ($n \geq 30$) and the population variance is known.
- *In both cases, we expect the data to be normally distributed*

Hypothesis Testing

- statistical method for inferring population parameters based on sample data. It provides a structured approach for evaluating claims or assumptions about a population using empirical evidence.
- **The null hypothesis (H_0)** is a statement of no effect, difference, or relationship. It represents the status quo or the current understanding.
- **The alternative hypothesis (H_1)** is a statement that contradicts the null hypothesis. It represents the claim or the new understanding that the researcher wants to prove.

example

to determine if a new teaching method improves student test scores. You might form the following hypotheses:

- **Null hypothesis (H_0)**: The new teaching method has no effect on student test scores.
- **Alternative hypothesis (H_1)**: The new teaching method improves student test scores.
- *Hypothesis testing involves collecting sample data, calculating test statistics, and determining the probability of observing such results if the null hypothesis is true.*
- *Based on this probability, we can decide whether to reject the null hypothesis in favor of the alternative or fail to reject it.*

t-test

A t-test is a statistical test used to determine whether there is a significant difference between the means of two groups or between a sample mean and a known value. It is particularly useful when dealing with small sample sizes or when the population standard deviation is unknown.

- The t-test statistic for a one sample t-test is calculated using the formula:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

where:

•$\bar{X}$ is the sample mean

•μ is the population mean (or the mean of the comparison group)

•s is the sample standard deviation, and

•n is the sample size.

Types of t-tests

Each compares means under different conditions:

- **One-Sample t-test**: This test compares the mean of a single sample to a known value or population mean. It determines if the sample mean significantly deviates from a specific benchmark.
- For example, we can use a one-sample t-test to evaluate whether the average test score of a small class differs from the national average.
- **Independent Two-Sample t-test**: This test compares the means of two independent groups to determine if there is a statistically significant difference between them. It is commonly used in experiments where two groups undergo different treatments or conditions.
- For instance, we could use an independent two-sample t-test to compare test scores between students taught using two different teaching methods to see if one method is more effective.
- **Paired t-test**: This test compares means from the same group at different times or under different conditions. It evaluates whether there is a significant change within the same group after an intervention or over time.
- An example is measuring student performance before and after implementing a new teaching strategy to assess its impact.

Assumptions of the t-test

The t-test relies on certain assumptions to provide valid results:

- **Normality of the Data**: The t-test assumes that the data in each group are approximately normally distributed. This is especially important when dealing with small sample sizes. If the data are not normally distributed, the t-test results may be unreliable.
- **Homogeneity of Variances**: For an independent two-sample t-test, the variances of the two groups being compared are assumed to be equal. This assumption ensures that the t-test correctly accounts for variability within each group. If the variances are not equal, it can affect the accuracy of the test.
- **Independence of Observations**: The observations within each group should be independent. This means that the value of one observation should not influence or be related to the value of another observation. Violation of this assumption can lead to incorrect conclusions.

Z-test

A Z-test is a statistical test used to determine whether there is a significant difference between the sample mean and the population mean or between the means of two groups when the population variance is known, and the sample size is large.

- It is primarily used when the sample size exceeds 30, allowing the use of the normal distribution to approximate the distribution of the test statistic.

$$z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Types of Z-tests

- **One-Sample Z-test**: This test compares the mean of a single sample to a known population mean. It is used when you want to assess whether the sample mean significantly deviates from the population mean, assuming the population variance is known. For example, a one-sample z-test might be used to determine if the average height of a group of more than 30 people differs from the known national average height.

- **Two-Sample Z-test**: This test compares the means of two independent samples to determine if there is a significant difference between them. It is used when both samples are large and the population variances are known. An example of this would be comparing the average test scores of students from two different schools to see if there is a significant difference in performance between the two schools.

- **Proportion Z-test**: This test compares the proportion of a certain characteristic in a sample to a known population proportion or between two sample proportions. It is used to evaluate whether the observed proportion in the sample significantly differs from what is expected based on the population proportion. For instance, a proportion Z-test might be used to compare the proportion of voters favoring a particular candidate in a sample to the proportion observed in previous elections.

Assumptions of the Z-test

- **Known Population Variance**: The Z-test assumes that the population variance is known. This is a key distinction from the t-test, where the population variance is typically unknown. The known variance allows for using the z-distribution to assess the significance of the test statistic.
- **Large Sample Size**: The Z-test assumes a large sample size, typically greater than 30. With larger samples, the sampling distribution of the sample mean approaches a normal distribution, even if the original data are not normally distributed, according to the Central Limit Theorem.
- **Normal Distribution of the Population**: The data are assumed to be drawn from a normally distributed population. This assumption is less critical for large samples but still important when the sample size is moderate.

Practical applications and use cases

- **t-test**: The t-test is commonly used in small-sample studies, such as pilot studies, where the population variance is unknown. Examples include comparing the effectiveness of two treatments in a small group or assessing changes within the same group over time.
- **Z-test**: The Z-test is used in large-sample studies or when dealing with well-established populations where the variance is known. It is often applied in quality control, survey analysis, and large-scale experimental studies.

Here is table with the key differences:

Aspect	T-Test	Z-Test
Sample Size	Small (n < 30)	Large (n > 30)
Population Variance	Unknown	Known
Distribution Assumption (for the test statistic)	T-distribution - heavier tails	Normal distribution
Application	Small-sample studies, unknown variance	Large-sample studies, known variance

One Sample Z test

Suppose a company claims that their new smartphone has an average battery life of 12 hours. A consumer group tests 100 phones and finds an average battery life of 11.8 hours with a known population standard deviation of 0.5 hours.

Step 1: Hypotheses:

H_0: $\mu=12$

H_1: $\mu\neq12$

$$z = \frac{11.8-12}{\frac{0.5}{\sqrt{100}}} = -4$$

- **Step2: Calculate the Z-Score**:
  we can calculate Z-score using the formula:
- where $\bar{x}=11.8$,$\mu=12$, $\sigma=0.5$ and $n=100$
- **Step3: Decision**
  Since $|Z|=4>1.96$ (critical value for $\alpha=0.05$) we reject H_0 indicate significant evidence against the company's claim.

The Z-Table

You must calculate your z-statistic using the formula then compare it at the difference significance levels

If you have a two tailed example you compare your z value to the correct significance level on $z_{1-\alpha/2}$. For example: if your two tailed z statistic was 1.98, for the 5% level you would compare it to 1.96 as it is greater this is significant!

Significance level	10%	5%	1%
$z_{1-\alpha}$	1.28	1.645	2.33
$z_{1-\alpha/2}$	1.645	1.96	2.58

These are the most common significance levels, although there are others!

If you have a one tailed example you compare your z value to the correct significance level on $z_{1-\alpha}$

Perform z-test

- **Problem 2:** A researcher wants to compare the effectiveness of two different medications for reducing blood pressure. Medication A is tested on 50 patients, resulting in a mean reduction of 15 mmHg with a standard deviation of 3 mmHg. Medication B is tested on 60 patients, resulting in a mean reduction of 13 mmHg with a standard deviation of 4 mmHg. At a 1% significance level, is there a significant difference between the two medications?

A chi-square distribution

A chi-square distribution is an asymmetrical family of distributions. A chi-square distribution with v degrees of freedom is the distribution of the sum of the squares of v independent standard normally distributed random variables. Intuitively, chi-square distributions take only non-negative random variables.

A chi-square distribution is used to test the variance of a population that is distributed normally.

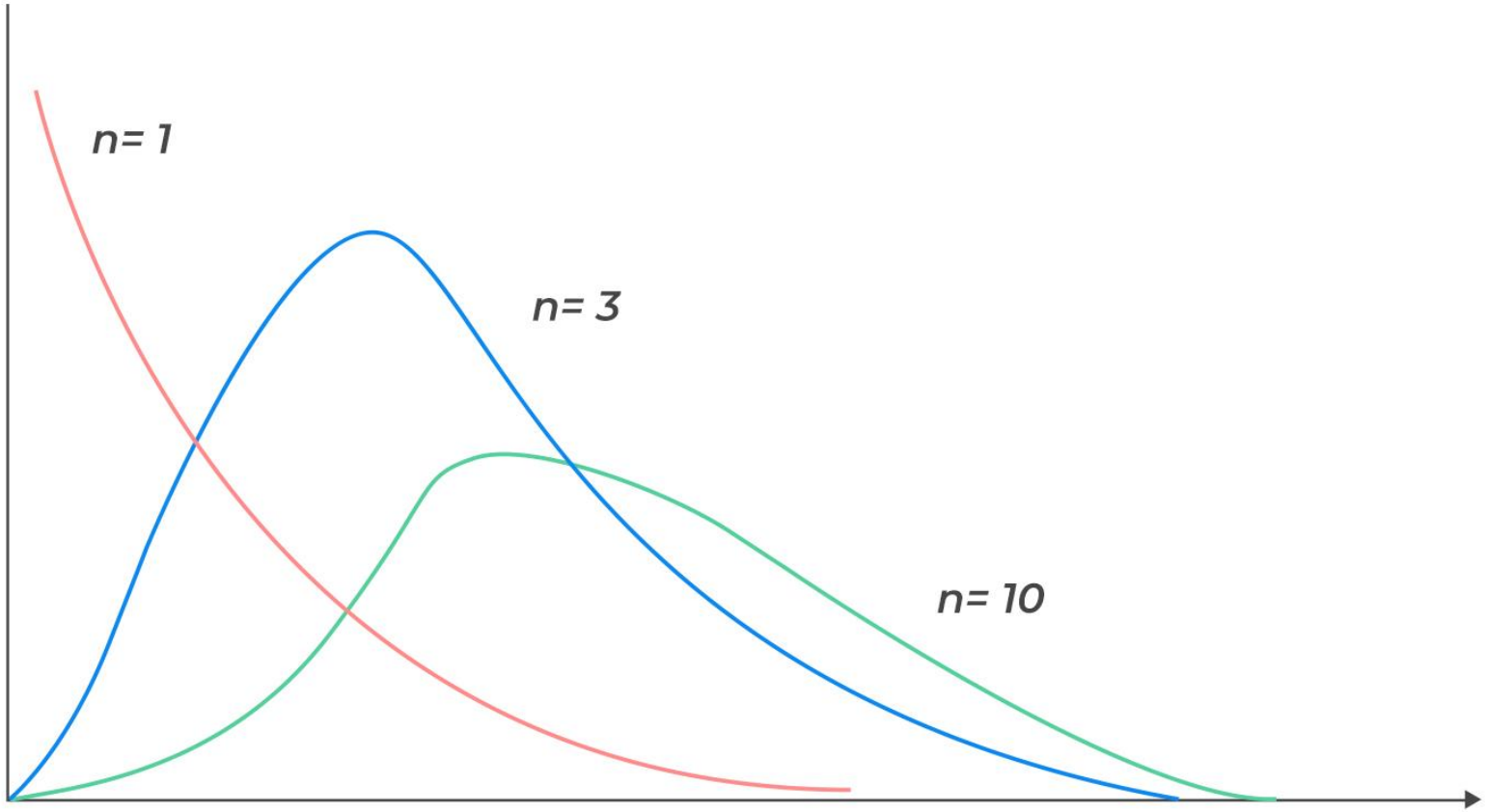
the following are the properties of a chi-square distribution:

- A chi-square distribution is a non-symmetrical distribution (skewed to the right).
- A chi-square distribution is defined by one parameter: Degrees of freedom (df), v=n–1.
- A chi-square distribution is the sum of the squares of k independent standard normally distributed random variables. Hence, it is a non-negative distribution.

A chi-square distribution

- For each degree of freedom, there are different chi-square distributions.
- The shape of a chi-square distribution changes with the change in the degrees of freedom. The more the degrees of freedom increase, the more the distribution assumes the shape of a standard normal distribution.

Chi-square Distribution



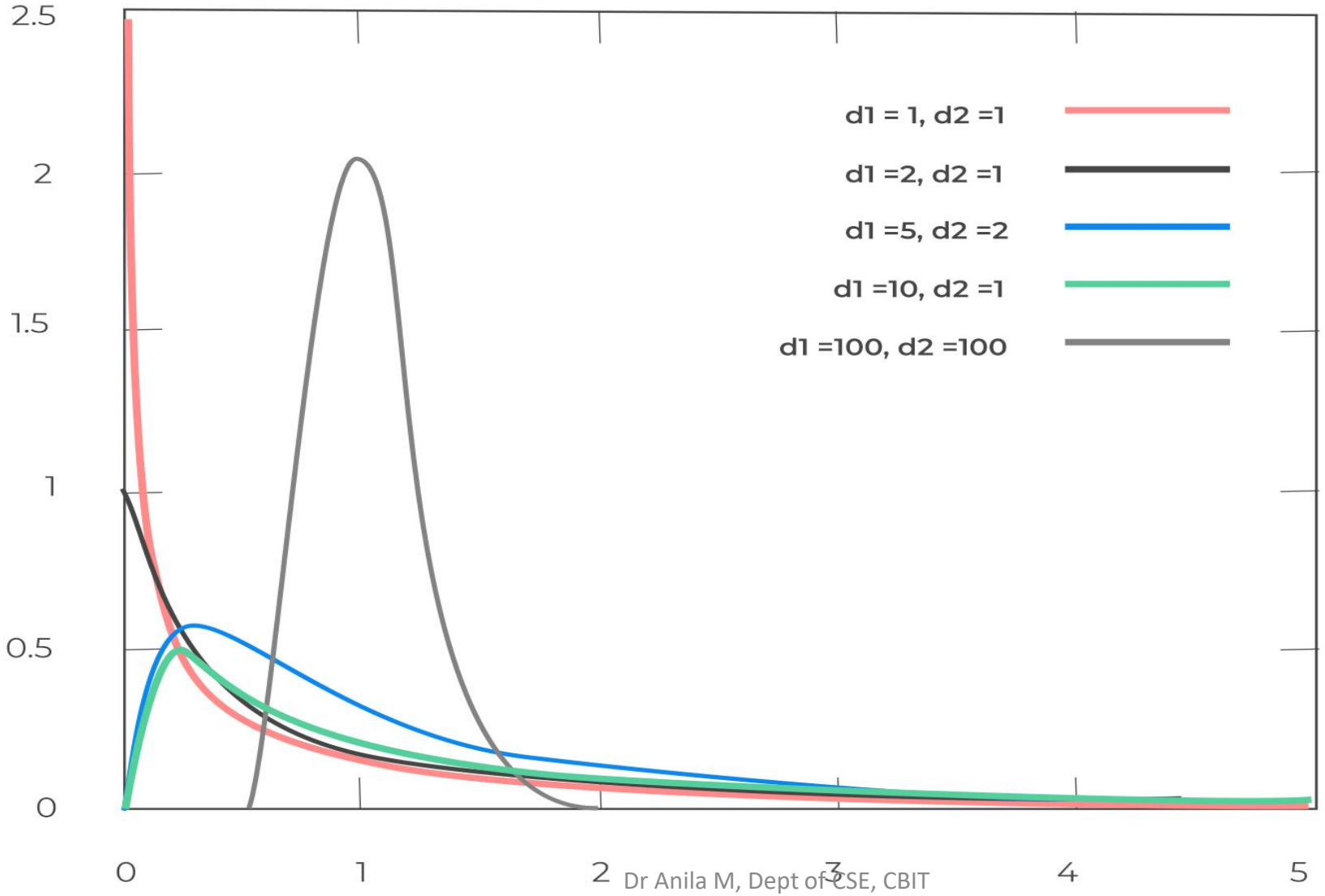
F-distribution

An F-distribution is used to test the equality of variances of two normally distributed populations from two independent random samples.

The following are the properties of an F-distribution:

- An F-distribution is an asymmetrical distribution (skewed to the right).
- An F-distribution is defined by two parameters, i.e., degrees of freedom of the numerator (m) and degrees of freedom of the denominator (n).
- Like a chi-square distribution, an F-distribution can only have positive values.
- As the degrees of freedom for the numerator and the denominator increase, the F-distribution approximates the normal distribution.

F-Distribution



Relationship between the Chi-square and F-distributions

The F-distribution is the ratio of two chi-square distributions with degrees of freedom mm and n, respectively, where each chi-square has first been divided by its degrees of freedom, i.e.,

$$F = \frac{\left(\frac{\chi_1^2}{m}\right)}{\left(\frac{\chi_2^2}{n}\right)}$$

Where m is the numerator degrees of freedom and n is the denominator degrees of freedom.

example

- F and chi-squared statistics are really the same thing in that, after a normalization, chi-squared is the limiting distribution of the F as the denominator degrees of freedom goes to infinity. The normalization is

- chi-squared = (numerator degrees of freedom) * F
- For instance, if you tell me that you have an F(2,71) = 2.05, the corresponding chi-squared is 2 * 2.05 = 4.1 and, by the way, the tail probabilities are virtually the same:

- F(2,71) = 2.05 p = .1363
- chi2(2) = 4.1 p = .1287

ANOVA

ANOVA is a statistical test used to determine if there are significant differences between the means of two or more groups. It analyzes the variance within and between groups to assess whether the differences observed are due to random chance or actual group differences.

- **ANOVA is commonly used when you have a continuous dependent variable and one or more categorical independent variables with multiple levels.** The test compares the means across the groups and calculates an F-statistic and p-value to determine if the differences are statistically significant.
- *Types: one-way ANOVA, two-way ANOVA, and factorial ANOVA*

Chi-Square test

- **Chi-Square tests are often used when you have categorical data and want to determine if there is a relationship between two variables.** It is commonly used in fields such as social sciences, biology, market research, and quality control.

Types:

1. Chi-Square test for independence (to test if variables are independent) and
2. the Chi-Square goodness-of-fit test (to test if observed frequencies fit an expected distribution).

ANOVA or Chi-square test?

- **ANOVA is used to compare means across multiple groups with continuous dependent variables and categorical independent variables. On the other hand, Chi-Square tests assess the association or independence between categorical variables.** The choice between ANOVA and Chi-Square depends on the nature of the variables you are analyzing and the research question you want to answer.

When to use?

Use ANOVA when:

1. You have a continuous dependent variable (e.g., height, weight, test scores) and one or more categorical independent variables (e.g., treatment groups, age groups, education levels).
2. You want to compare means across two or more groups.
3. You are interested in assessing the statistical significance of group differences.
4. You want to determine if the variation within groups is greater than the variation between groups.

Example**: you might use ANOVA to analyze the effects of different teaching methods (independent variable) on student test scores (dependent variable) across multiple classes.**

When to use?

Use Chi-Square when:

1. You have categorical variables (e.g., gender, occupation, preferences) and want to test for associations or independence between them.
2. You want to determine if the observed frequencies of categories differ significantly from the expected frequencies.
3. You are interested in analyzing the relationship between two or more categorical variables.

NOTE

- use Chi-Square to examine whether there is an association between smoking habits (variable 1: smoker, non-smoker) and the incidence of lung cancer (variable 2: diagnosed with lung cancer, not diagnosed) in a population.